

**Unit ID: 679b0aa7aaf593b2823f310e**

## **Unit 2 - How Smart Bins Work**

### **PDF Documents**

**PDF Document 1: 1738214054103-Chapter 6 - Unit 2 - How Smart Bins Work.pdf**

This PDF document is included in the unit materials.

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## UNIT 2 HOW SMART BINS WORK

### ● Introduction to Smart Bins

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Smart bins are special trash bins equipped with sensors that can detect how much garbage is inside. These sensors can tell when the bin is almost full and automatically send a signal to the waste management system to schedule a pickup. This smart technology helps keep areas clean without overflow.

Unlike regular trash bins, which require manual checking to see if they need to be emptied, smart bins handle this automatically. They make managing waste more efficient because they provide real-time updates on their status, ensuring that they are only emptied when necessary. This not only saves time but also reduces the cost and environmental impact of waste collection.



## HOW SMART BINS USE SENSORS

Smart bins often use a variety of sensors to determine how full they are:

**ULTRASONIC SENSORS** These sensors measure the distance from the top of the bin to the trash surface. They send out ultrasonic waves, and when these waves bounce back from the trash, the sensor calculates how full the bin is based on the time it takes for the waves to return.

**WEIGHT SENSORS** Installed at the bottom of the bin, these sensors measure the weight of the trash accumulated. As the bin fills up, the weight increases, and the sensor records this change to indicate how full the bin is.

**INFRARED SENSORS** These work by detecting the heat or the light reflected from the contents of the bin. They can gauge how close the trash is to the top of the bin, providing another method to measure fullness.

## HOW SMART BINS USE SENSORS



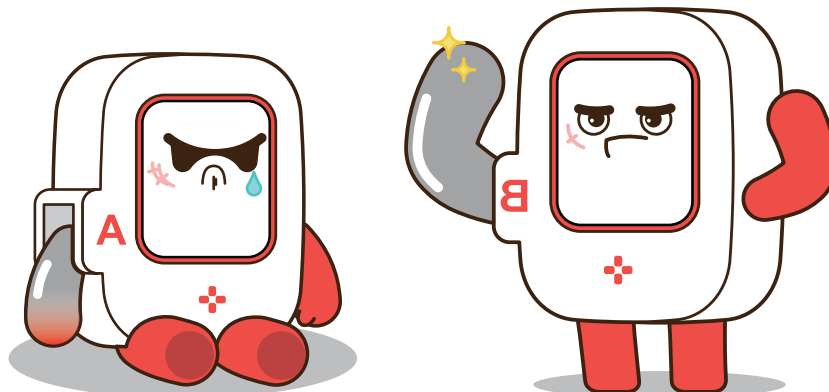
Smart bins use Wi-Fi or cellular networks to communicate their status to a central waste management system. When sensors in the bins detect that they are nearing full capacity, they send this information over the internet using these networks. The central system receives these updates and can then efficiently plan waste

collection routes and schedules based on real-time data from multiple bins. This connectivity ensures that waste collection is timely, preventing overflow and optimizing the process for cities and municipalities.

## BENEFITS OF SMART BINS



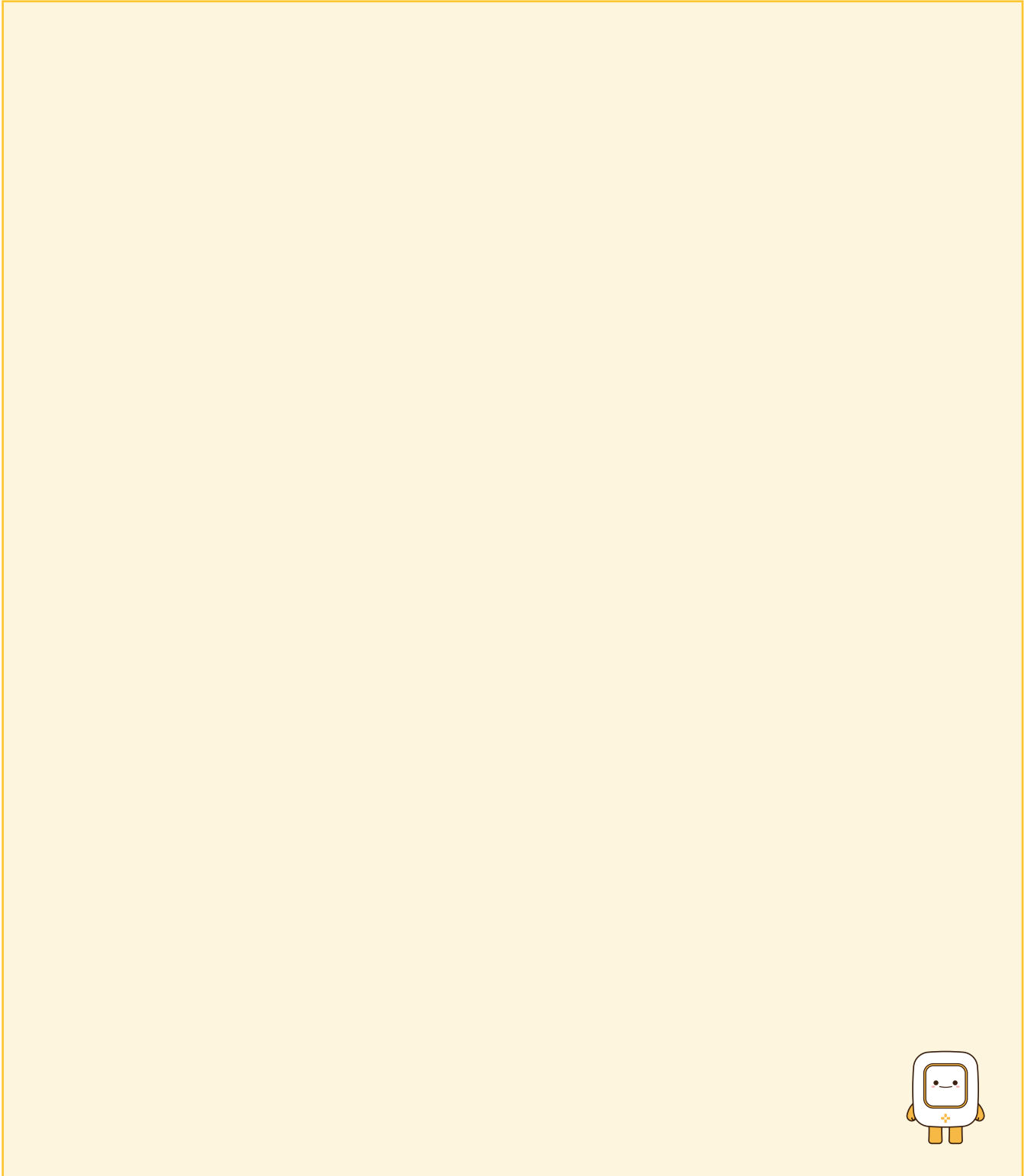
Smart bins contribute significantly to urban cleanliness and efficiency. By using sensors to monitor fill levels and alert waste management systems when nearing capacity, they prevent overflow of trash, which can lead to litter-strewn streets and attract pests. Additionally, the data-driven approach allows for more efficient routing of collection trucks, ensuring that only the bins needing service are visited, which conserves fuel and reduces traffic congestion. This smarter allocation of resources not only keeps the streets cleaner but also enhances the overall sustainability of city operations.





## **Activity:** \_\_\_\_\_

Draw a picture of a regular trash bin and a smart bin and label the parts that might be different.



 **Activity:** 

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Role-play activity where one student is a smart bin, one is the central system, and others are trash collectors. The smart bin student sends a message when full, and trash collectors respond.